



August 8, 2025

Re: Response to Reviewer Comments
MS# 2025-07962

Dear Dr. Berenbaum,

We thank the editor for the opportunity to revise our manuscript, “Evolution under vancomycin selection drives divergent collateral sensitivity patterns in *Staphylococcus aureus*.” We also appreciate the thoughtful and positive evaluations and suggestions provided by the two external reviewers. Their feedback has improved the manuscript. In this letter, we respond to their comments below and highlight the changes made in our revised manuscript.

Reviewer 1:

This is a potentially interesting study that describes mutations arising in *S. aureus* in the presence of vancomycin, in an evolutionary experiment. Different lineages of *S. aureus* from parallel experiments acquire different mutations, and there is a divergence among those carrying *walKR*; and *rpsU*. WalKR is a two-component regulator of cell wall biosynthesis, and mutations in this regulon lead to a thicker cell wall and emergence of resistance to vancomycin both in vitro and in the clinic. RpsU is a ribosome-related stress response regulator, and clinical isolates resistant to vancomycin do not have mutations in this locus. Apart from these two mutations, each evolved strain carries a large number of additional mutations, their possible role in vancomycin resistance is unclear. The authors also examine collateral sensitivity of these vancomycin resistant mutants and find increased susceptibility to clindamycin and nafcillin. This study is well-executed, but its significance is not obvious, given that only the *walKR* mutants are clinically significant.

We appreciate your compliment regarding the execution of our study, and you raise an important point about its clinical significance.

Although WalKR regulon variants are well-represented in the VISA literature, independent studies document additional loci that emerge under vancomycin therapy in clinical infections. For example, Basco et al. (2019) sequenced seven longitudinal bloodstream isolates from a patient treated with vancomycin. Mutations in *rpsU* arose and co-occurred with rising vancomycin MICs and hVISA phenotypes, confirming their in vivo contribution to reduced vancomycin susceptibility. Moreover, Hafer et al. (2012) identified mutations in the envelope-stress regulators *vraS* and *liaF* in several clinical VISA isolates, also confirming their clinical importance. We cited these studies in our original paper; however, in the case of Basco et al., we did not explicitly mention that it was performed on clinical isolates. Our discussion would benefit from further clarification. We therefore expanded the paragraph on pages 5 and 6 (Lines 160–165 in the revised manuscript), explicitly tying the vancomycin-associated mutations in our lines with those found in clinical isolates (we bold the additional sentences):

Weighted elastic-net logistic regression identified several genes associated with vancomycin exposure. First, *vraS* and *liaF* genes were the most frequently mutated in our lines (Fig. 2B). They occurred in seven out of 18 VISA populations but were absent in all 87 control lines (Fig. 2C). LiaF and VraS regulate the cell membrane stress response (Suntharalingam et al., 2009; Galbusera et al., 2011) and are prevalent in clinical VISA isolates (Hafer et al., 2012). Second, *rpsU* mutations targeting the 30S ribosomal subunit occurred in six VISA lines, and they were also absent in any control population. Ribosomal perturbation can activate stress response in some bacteria. *Listeria monocytogenes* *rpsU* mutants exhibit upregulated σ^B -mediated stress genes and slower growth, trading fitness for broad stress resistance (Koomen et al., 2024). In our lines, *rpsU* mutations may similarly impact global regulation, resulting in an elevated cell membrane stress response and reduced cell wall turnover. ***rpsU* mutations have also been implicated in increased vancomycin resistance in clinical infections (Heidarian et al., 2024; Basco et al., 2019).** For example, Basco and colleagues sequenced longitudinal isolates from a bacteremic patient treated with vancomycin (2019). Mutations in *rpsU* arose within five days of treatment, and these variants were associated with the VISA phenotype, **confirming their in vivo contribution to reduced vancomycin susceptibility.** Third, *yycH*, *yycI*, *walK*, and *atl* mutations repeatedly occurred in our evolved VISA lines. These genes comprise the WalKR regulon, an essential cell-wall metabolic pathway in *S. aureus* (Fig. 2D) (Dubrac et al., 2007; Gajdiss et al., 2020). Loss-of-function mutations in any of these regulon genes cause increased cell wall teichoic acid content, resulting in a thicker and more resistant cell wall against vancomycin and daptomycin (Gajdiss et al., 2020; Sulaiman et al., 2022). This finding is consistent with our results and an earlier study, which showed that mutations in *walKR* cause increased resistance to these antibiotics in clinical VISA strains (Howden et al., 2011). Together, these results illustrate that vancomycin selection promoted strong genomic parallelism by driving repeated mutations in a limited set of genes primarily involved in cell wall metabolism and general stress response, raising the question of whether evolutionary constraints influenced these adaptive trajectories.

Another issue is novelty, the authors site a recent study, in this Journal, that conducted a similar in vitro evolutionary experiment, adapting *S. aureus* to vancomycin (Fait et al., 2002, ref. 25). In the Fait study, more mutations obtained in vitro corresponded to those in clinical isolates (*vraT*, *graS*, *vraFG*, *walKR*, and *rpoBCD*), and there were new ones as well (SAUSA300_RS04225, *ssaA*, *pitAR*, and *sagB*). The new mutations observed in both this and Fait study are probably not present in clinical isolates because they have a considerable fitness cost in vivo, but not in vitro, underscoring the concern regarding their relevance to the clinical manifestation of staphylococcal infections and vancomycin resistance.

While we acknowledge that the clinical prevalence of adaptive mutations is an important measure of a study's relevance, it is not the sole indicator of its significance. As previously described, many of the mutations observed in both studies occur in vivo. Nonetheless, the novelty of our research lies in what it reveals about the fundamental evolutionary strategies available to *S. aureus* when exposed to vancomycin, and how those strategies shape its subsequent susceptibilities to first-line drugs. Our key findings demonstrate that *S. aureus* can follow at least two distinct adaptive routes. One strategy involves modifying the cell wall, mediated by mutations in the WalKR regulon. A second, alternative strategy involves a broader reprogramming of translational stress-response pathways, mediated by *rpsU* mutations. The divergence between these two pathways was stark; we observed near complete mutual exclusivity between mutations in the WalKR regulon and those in *rpsU*. This strong negative association suggests that these are not random occurrences but represent

two competing, viable evolutionary solutions to vancomycin stress. We further demonstrated that these pathways led to divergent collateral responses to clinically used first-line drugs. Therefore, we argue that our study is novel because it highlights a crucial and sometimes underappreciated aspect of bacterial evolution that is of clear interest to clinicians: evolutionary routes are sometimes shaped by prior mutations in a historically contingent manner, and this contingency can have clinical consequences.

Additionally, in several instances throughout the Results and Discussion section, we highlighted the limitations of our study and outlined potential future research avenues that could enhance its clinical applicability. We have now revised the manuscript in several places to provide additional clarity (changes shown in bold).

Lines 272 and 273 (revised manuscript):

Our findings, however, may depend on the specific environmental conditions under which antibiotic susceptibilities were assessed. For instance, Machado and colleagues (2021) evolved MRSA populations under different media conditions — cation-adjusted Mueller-Hinton broth (CA-MHB) and Roswell Park Memorial Institute (RPMI) medium — to simulate bacteriological and physiological conditions, respectively. Their results indicated trade-offs between environmental conditions, with lines evolved in CA-MHB showing increased vancomycin resistance in RPMI, but not vice versa. This result suggests that the susceptibility profiles observed in our VISA populations, measured in CA-MHB, may also differ under physiologically relevant conditions. Additionally, Machado and colleagues identified mutations in *walK*, *yycH*, and *vraS*, which are consistent with our study, though *liaF* mutations were notably absent in their lines. This discrepancy could reflect differences in genetic background, as we evolved MSSA populations, whereas their study used MRSA. Thus, genetic background and environmental context together shape collateral responses, reinforcing the clinical relevance of considering both factors when interpreting experimental evolution studies **and underscoring the need for future in vivo investigation.**

Lines 350–352 (revised manuscript):

Third, **we identified key mutations in our populations (e.g., *walKR*, *rpsU*, *liaF*, and *vraS*) that are also present in clinical VISA isolates, suggesting that our genetic findings extend beyond laboratory conditions.** However, we acknowledge that our experimental design, which involved progressively increasing vancomycin concentrations, differs from the dosing regimens typically used in clinical settings. This approach may have influenced the evolutionary trajectories observed in our study, potentially leading to the accumulation of mutations distinct from those that would occur under constant therapeutic doses. For example, Oz and colleagues showed that when *E. coli* lines evolve antibiotic resistance under strong selection, they exhibit broader collateral resistance and have more mutations in drug pathway-specific genes than populations evolved under weak selection (2014). While our approach allowed us to capture a range of adaptive outcomes, it might overrepresent some variants more than others. Further studies under higher, clinically relevant doses (and over shorter durations) are warranted to understand the dynamics of resistance evolution in therapeutic contexts.

Reviewer 2:

This is an elegant study of the effects of diverse genetic pathways to vancomycin adaptation by *S. aureus* can influence its collateral effects on other resistance traits and hence its potential treatment. Studies like this are growing in number to improve forecasting of treatment outcomes and to anticipate next steps beyond initial treatment failure. The evolve-and-resequence approach has been shown to be rigorous and to demonstrate the best genetic pathways that balance fitness and resistance. I also applaud the effort to describe interactions between genetic pathways, ie how do certain first-step mutations either favor or disfavor second steps.

Thank you for these positive and encouraging words about our study. We, too, find the potential epistatic interactions between pathways fascinating, and this is a topic we plan to explore in the future.

The major finding appears to be the relationship between collateral sensitivity and adaptive pathway, and the scoring of these CS effects. This is a promising first step to quantify consequences of resistance evolution. To take this study to the next level it would be quite useful to test whether lines predicted have vulnerabilities are indeed vulnerable during treatment, and or how durable their susceptibilities are during exposure to a new second or third drug.

We appreciate your kind remarks about the promise of our study as a first step in characterizing the consequences of resistance evolution. We agree that the studies you mentioned are logical next steps, and we have already begun a follow-up project along these lines. Susceptibility measurements are “static,” only capturing a snapshot of a population’s current phenotypic state without accounting for its future potential to evolve resistance. As we have shown, prior mutations can limit a population’s ability to adapt in new directions under stressful conditions (Card et al., 2019; Card et al., 2021). Similarly, a pathogen’s evolutionary history under one antibiotic may alter its adaptability to subsequent drugs, limiting or enhancing their effectiveness. In our current work, we are investigating this possibility by establishing replicate *S. aureus* populations from clones that exhibit increased first-line drug sensitivity. We are propagating these lines under increasing concentrations of the same drug, using the resulting phenotypic and genomic results to evaluate how a strain’s genetic background impacts its potential to evolve resistance. We ask whether lines can overcome their initial sensitivity through further evolution? And for those that cannot, which mutations are responsible for this evolutionary constraint?

I have a few concerns about the overall analysis that limit my current enthusiasm. Most have to do with a need for more detail about the bioinformatic analyses. I could not find a table of the primary mutation calls, which must be shared, and this limited my inferences.

This is a fair concern. In our original paper, we provided a table of mutation calls in our associated GitHub repository, along with its URL, in the Data Availability statement and references. We now include these data in the Supplementary Information (Dataset S1). Moreover, we have now deposited all *breseq* data, including raw HTML outputs, FASTA files, Variant Call Format (VCF) files, Binary Alignment Map (BAM) files, and GenomeDiff files for all 105 populations in Dryad (DOI: 10.5061/dryad.qnk98sfw2).

Further, the original premise of the 87 control lines, however elegant, was not realized (Lines 114-121). What did we learn about the "ensemble of outcomes" that differ from the experimental group?

Thank you for this comment. We agree that more should have been said about the genetic responses

of the 87 control lines in our original paper. To address this point, we have added a paragraph (and three additional references) to the Results and Discussion, as follows:

Lines 175–188 (revised manuscript):

Conversely, the logistic regression highlighted alleles that arose only in the antibiotic-free controls, representing adaptation to the general laboratory conditions, distinct from the selective pressure of vancomycin. Most prominent were *arlR* and its downstream effector *mgrA*, which accumulated in more than one-quarter of the control lines (Fig. 2C). Together, this two-component regulator and transcription factor modulate more than 100 genes involved in virulence, metal transport, adhesin production, and biofilm formation, functions likely tuned by nutrient availability (Crosby et al., 2020). Additional mutations affected the RNA-polymerase α subunit RpoA; the penicillin-binding protein PbpB, possibly indicating that these lines evolved reduced penicillin susceptibility; pyruvate kinase Pyk, an essential metabolic enzyme that plays a role in biofilm formation (Vasu et al., 2014); and polynucleotide phosphorylase (PNP), an enzyme involved in RNA metabolism and virulence regulation (Joo et al., 2016). Collectively, these genomic responses indicate that, in the absence of vancomycin, selection favored regulatory rewiring that optimizes core metabolism and surface remodeling, in contrast to the stress-response and cell-wall mutations selected during vancomycin exposure.

Major concern:

L142: 4921 reported mutations seems highly excessive given results from many other evolution experiments and is indicative of either high mutation rate, poor quality control of inferred variants, or both. This amounts to roughly 50 mutations per line, of uncertain distribution, and unclear relationship between genome region and coverage. Even with population sequencing where you include mutations down to 5% frequency, it's likely that most of these are false positive. If I'm wrong, more evidence of the variant calling quality is required because raw breseq analyses on *S. aureus* genomes always returns many false positives.

You raise an important point regarding the number of observed mutations. We agree that 4,921 variants is a large number, but we argue it is not excessive and is, in fact, much lower than theoretically expected given our experimental design. The high number reflects two main factors: (1) our sequencing of whole-population samples rather than single clones, and (2) the substantial number of mutations that accumulate during serial batch culture.

For example, the expected number of new mutations generated during population outgrowth is a product of the per-cell mutation rate (μ) and the number of cell divisions ($N_f - N_0$). Given a starting population (N_0) of 5×10^5 cells, a final population (N_f) of 2×10^8 cells, and a μ of 2.76×10^{-4} mutations per cell per generation, the number of mutations expected each day is approximately 55,000. Because our populations evolved for an average of 9 days, the total number of mutations generated per population was roughly **495,000**.

The ~50 mutations we report per line (at a frequency of $\geq 5\%$) represent only ~0.01% of this total mutational input. This small fraction illustrates the powerful filtering effects of selection and genetic drift, where the vast majority of new mutations are either deleterious or lost by chance before reaching a detectable frequency.

Additionally, if most variants were false positives generated during library preparation or sequencing, then the average number of mutations per line should be independent of the selective environment and roughly equivalent between our experimental and control groups. Our results do not support this hypothesis.

On average, each vancomycin-selected experimental line accumulated 14.6 mutations, whereas each control line grown without the drug accumulated 53.8 mutations. This significant difference is a classic signature of the intense evolutionary pressure imposed by the antibiotic. Under strong selection, populations undergo selective sweeps, where a few highly beneficial mutations rise to high frequency, purging other variants from the population. In contrast, the control populations experience more relaxed selection, allowing a greater diversity of mutations to accumulate and rise to detectable frequencies. This finding suggests that most of our reported variants are tied to these evolutionary processes, not to random technical artifacts.

Although the expected number of mutations and the genomic results indicate that *most observed variants in our lines are unlikely to be false positives*, we acknowledge that *some* may be. To evaluate this possibility, we performed two analytical approaches: binomial tests and weighted elastic-net logistic regression.

For both approaches, we began by curating structural variants, including indels less than 50 bp, large deletions, and duplications identified by new junction (JC) evidence. We relied on a distinct set of quality metrics that assess the integrity of split-read alignments. A putative junction was considered a true positive only if it satisfied both of the following criteria. First, the junction had to be supported by reads that aligned to different starting positions in the reference sequence, as indicated by a position-hash score of three or higher. This criterion filters out artifacts arising from PCR. Second, the junction had to be supported by reads mapping to both strands, thereby mitigating artifacts common to a single strand that can occur during library preparation or sequencing. All putative structural variants passing these filters were visually inspected to confirm the supporting read alignments. If a variant did not satisfy either criterion, we excluded it as a false positive. Additionally, we manually curated SNPs and 1-bp indels supported by read alignment (RA) evidence that occurred at the same genomic position across replicate populations. If the mutation occurred in a homopolymer tract or a tandem repeat region, we classified it as a false positive. Lastly, we excluded mutations within multicopy elements, such as ribosomal RNA operons and synthetases responsible for charging tRNA. **One hundred and eighty-six variants out of 4,942 were excluded in this step.**

1. Binomial tests: For base substitutions and small indel (1-bp) events supported by RA evidence, we performed a binomial test to assess the significance of each call. We reasoned that for a “true” variant, the number of reads supporting the alternative allele must be significantly greater than what would be expected from sequencing error alone. To this end, we modeled the occurrence of errors at each genomic position using a binomial distribution. The probability of success in a single trial (i.e., observing the alternative base due to error) was defined by the empirically derived Phred-scaled quality score for that specific base in each read. For each putative variant, we then calculated a p -value representing the probability of observing at least the given number of alternative-allele reads by chance. Variants with a p -value less than a Bonferroni-corrected threshold of 0.05 were retained as true positives.

We do not include information about this analysis in our revised manuscript because we opted to report the results from the weighted logistic regression. However, to perform the binomial tests, we extracted read coverage for each variant locus. We agree with you that the relationship between genomic regions and coverage was unclear in our previous draft. Thus, we now include this information in our mutation list, along with the number of reads supporting the alternate allele (Dataset S1). The minimum read coverage was 5, and the maximum was 756, with an average of 65. Moreover, the median Phred quality score was 40, corresponding to an error rate of 10^{-4} .

Only a handful of RA-supported variants were false positives. Only six resulting p -values were non-significant and, therefore, classified as false positives. Five of those mutations were found in control populations, and the remaining mutation was identified in an experimental line, occurring in a gene encoding a hypothetical protein. The remaining 4,720 p -values were highly significant. If we assume *breseq*'s filters are 100% accurate, and those six variants are true positives, then there is a 99.88% concordance between *breseq*'s filtering algorithms and the binomial analysis. The six variants in question do not change the outcome of any downstream analysis in our study.

2. Weighted elastic-net logistic regression: To identify genes associated with vancomycin selection, we enhanced our previous approach by implementing a weighted elastic-net logistic regression. This updated model leverages the *glmnet* R package and improves upon our former method by directly incorporating variant confidence scores into the feature selection process.

Our refined workflow is as follows:

- (a) Model Construction and Weight Assignment: As before, we began with a set of qualifying mutations that occurred in at least three populations (resulting in 484 gene-level features). The key enhancement in our new model is the assignment of a specific regularization penalty to each feature based on its *breseq* confidence scores. In a discretized fashion, variants with low, medium, and high confidence were assigned weights of 10.0, 1.0, and 0.1, respectively. This discretization forces the model to penalize low-confidence variants more heavily, thereby prioritizing the selection of high-confidence, reliable mutations.
- (b) Feature Stability Analysis and Selection: To identify a robust set of informative features, we first determined the optimal model hyperparameters (α and λ) using k -fold cross-validation. We then performed a bootstrap analysis over 1,000 iterations, fitting the weighted elastic-net model with these fixed optimal parameters to a new bootstrap sample of the data in each iteration. Given the sparsity-inducing property of L1 regularization, we tracked the frequency of non-zero coefficients for each gene across all 1,000 runs. Mutations that exhibited non-zero coefficients in at least 10% of the iterations were retained, constituting the final set of stable, informative features for subsequent Bayesian latent class analysis (BLCA).

Finally, we computed the 95% confidence intervals for each selected gene's coefficient and exponentiated the endpoints to obtain the odds ratio, allowing us to determine the direction and strength of its association with the selective condition. In summary, this updated methodology is more rigorous as it formally uses the variant confidence scores to guide the feature selection process, leading to a more robust and reliable set of candidate genes.

Overall, we did not observe a significant difference in the results from this updated regression and our previous analysis. Although variants in *mdlD*, *smc*, *fntB*, and *prfA* in several control populations had low confidence scores, they were not associated with experimental conditions, and thus, were in both weighted and unweighted models.

We have now revised the Methods section, by adding the following paragraphs:

Lines 451–473 (revised manuscript):

To identify genes associated with vancomycin selection, we implemented a weighted elastic-net logistic regression using the glmnet R package. This approach incorporates both L1 (LASSO) and L2 (Ridge) regularization to perform feature selection on a high-dimensional dataset. We included qualifying mutations that occurred in at least three populations, resulting in 484 gene-level features for the model.

A key feature of our model is the integration of variant-calling confidence scores to guide the regularization process. Each gene-level feature was assigned a specific penalty weight based on its underlying *breseq* confidence scores. This step was implemented in a discretized manner, where variants with low, medium, and high confidence were assigned regularization weights of 10.0, 1.0, and 0.1, respectively. This weighting scheme forces the model to apply a stronger penalty to low-confidence variants, thereby prioritizing the selection of features supported by high-quality data.

To identify a robust set of informative features, we first determined the optimal model hyperparameters (α and λ) via k-fold cross-validation. We then performed a bootstrap analysis over 1,000 iterations, fitting the weighted model with these optimal parameters to a new bootstrap sample of the data in each run. Given the sparsity-inducing property of L1 regularization, we tracked the frequency of non-zero coefficients for each gene across all iterations. Mutations exhibiting non-zero coefficients in at least 10% of the iterations were retained, constituting the final set of stable features for subsequent Bayesian latent class analysis (BLCA) (SI Methods). Finally, we computed the 95% confidence intervals for each selected gene's coefficient and exponentiated the endpoints to obtain the odds ratio, allowing us to determine the direction and strength of its association with the selection condition.

In summary, the binomial approach demonstrates that only a handful of variants supported by RA evidence are, indeed, false positives. Moreover, even after manually curating and excluding suspect JC-evidence variants and down-weighting low-confidence RA variants in our regression model, we did not observe any difference in analysis outcome between our previous paper draft and this version.

Fig 2: control lines, in agreement with the above comment, in my experience *ebh* mutations are often false positives given its repetitive sequence. please examine carefully, manually and/or by clone sequencing as appropriate

We manually inspected the 49 variants in *ebh* and you are correct: many of the variants occurred in repetitive sequences. We now classify these as false positives and exclude them from downstream analyses.

Fig 2: control lines also seems to indicate high parallelism of mutations in *tilS*, which is unusual since it is an essential gene associated with tRNA modification. This is either a chronic issue of false positives or a very curious finding, but this deserves more scrutiny and discussion.

You are correct. Upon a manual, visual investigation of these variants, we noted that they all occurred in a tandem repeat region. We now classify them as false positives and exclude them from downstream analyses.

Fig 2: The WalKR and RpsU major targets are unsurprising genetic targets based on the literature of evolved AMR. What distinguishes your findings from what has been reported,

from the perspectives of the specific alleles or evolved phenotypes?

Thank you for raising this important point. You are correct that WalKR and RpsU are known genetic targets in the evolution of vancomycin resistance. Our study builds on this knowledge to uncover the downstream consequences depending on which path is taken. Our primary finding is not simply that these genes are mutated, but that the evolutionary pathways are constrained. We show strong evidence for sign epistasis, where an adaptive mutation in WalKR largely precludes subsequent adaptation in RpsU, and vice-versa. This observation highlights a critical genetic interaction that has not been previously characterized. We then connect these alternative genetic paths to distinct, clinically relevant phenotypes. We show that WalKR mutants and RpsU mutants develop significantly different collateral resistance profiles when challenged with other antibiotics. This finding has direct implications for treatment, as the initial evolutionary step could determine the success or failure of subsequent therapies.

We acknowledge that the specific alleles and phenotypes observed might be a product of our experimental conditions, and further work is needed to confirm these dynamics in vivo. However, we emphasize that the principle of historical contingency itself is not an experimental artifact. It is a fundamental force in evolution that, as we show, can lead to unpredictable clinical outcomes.

What mutations if any may depend on MHB or RPMI media?

Thank you for raising this important point. We would like to clarify that our experiments were conducted solely in MHB, and our discussion of RPMI was meant to contextualize our findings and acknowledge the limitations of using a single bacteriological medium. Although we cannot answer your question from our data, we form the following hypothesis based on the known differences between the media and the findings of Machado et al.

A key difference between our findings and the Machado et al. study was the presence of *liaF* mutations in our populations. We propose that these mutations may be favored specifically in MHB. The rich composition of MHB could interact with the antibiotic to create a specific type of cell envelope stress that is effectively mitigated by mutations in the LiaFSR pathway. In the different ionic and nutritional environment of RPMI, this same stress may be absent, favoring alternative adaptive pathways (like those seen by Machado et al.).

We also speculate that mutations related to metabolism are highly likely to be media-dependent. RPMI mimics physiological conditions and is limited in certain nutrients compared to MHB. Therefore, evolution in RPMI might select for mutations in amino acid biosynthesis pathways or high-affinity nutrient transporters. Such mutations would offer little to no benefit in the nutrient-rich MHB and thus would not be selected for in our experimental conditions.

This speculation underscores the main point of that discussion section: that the observed evolutionary trajectory is an integration of multiple factors. The discrepancy in *liaF* mutations could be due to the media (MHB vs. RPMI), the genetic background (MSSA vs. MRSA), or an interaction between the two.